higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PLANT AND EQUIPMENT
NQF LEVEL 2

(12030052)

20 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use a pen with only BLUE or BLACK ink.
6. Write neatly and legibly.

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QUESTION 1: INTRODUCTION TO PLANT AND MACHINERY

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 Domesticating animals is to tame them so that they are no longer wild.

1.2 Mechanisation occurs when humans use machines to help them do physical work.

1.3 The medieval era started around the tenth century CE.

1.4 Overhead costs are the cost of running a project, such as electricity, labour and rental of equipment.

1.5 A contractor is a person or a company that has won a contract to work for another company.

(5 × 1) **[5]**

QUESTION 2: LEGISLATION AND SAFETY REGULATIONS

2.1 Government has passed certain laws or legislation that describes the importance of safety in the construction industry. The laws are passed down by means of descriptions.

List FIVE pieces of legislation that describe the importance of safety in the construction industry. (5)

2.2 State the main purpose of health and safety legislation and regulations in the construction industry. (3)

2.3 The Occupational Health and Safety Act (OHSA) 85 of 1993 is important to the construction industry, as it provides clear rules on safe working conditions for both employer and employee.

2.3.1 State TWO workplaces in which workers are excluded from this Act. (2)

2.3.2 List THREE types of workers that the Act includes. (3)

2.4 Write out the following abbreviations which are used in Plant and Equipment Level 2:

2.4.1 PPE

2.4.2 SEMS

(2 × 1) (2)

[15]

QUESTION 3: FIRST AID

3.1 First aid has three different levels of priority.

List and describe the THREE levels of priority in first aid. (2 × 3) (6)

3.2 Name TWO places at which a pulse can be checked on the body of an adult or a child. (2)

3.3 State TWO signs that you will observe in a patient when checking for shock. (2)

3.4 List the THREE degrees of burns. (3)

3.5 Define the term *sterile*. (2)

[15]

QUESTION 4: HAND TOOLS

4.1 List FIVE types of saws that will fall under the category of carpentry tools. (5)

4.2 State the function of a recovery magnet. (2)

4.3 A polygon is a figure with many sides – normally more than four sides or angles.

List THREE forms of a polygon. (3)

4.4 Explain the process used to mark out, cut and assemble plywood to create a simple shuttering that can be used as a concrete mould. (10)

4.5 Explain the procedure required to make a simple oilstone box for two pieces of timber using a plane and wooden chisel. (10)

[30]

QUESTION 5: LADDERS

5.1 State FIVE guidelines for placing a ladder correctly. (5)

5.2 Explain the dangers of over-reaching when working from a ladder. (5)

[10]

QUESTION 6: TRESTLES, SCAFFOLDING AND MOBILE TOWERS

6.1 Choose an item/word from COLUMN B to match a description in COLUMN A. Write only the letter (A–L) next to the question number (6.1.1–6.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
6.1.1	A certificate that the person in charge of erecting the scaffolding gives to the site supervisor	A	couplers
6.1.2	A platform that rests on a pair of trestles	B	strut
6.1.3	Joins pipes together to make the scaffolding	C	sole boards
6.1.4	Fits the bottom of a standard and makes it stable	D	cantilever
6.1.5	A thin piece of metal or wood that is used to strengthen the overall structure	E	non-compliance
6.1.6	The surface that will carry the weight of the scaffold	F	handover certificate
6.1.7	A projecting structure that is supported or attached at one end only	G	clearance certificate
6.1.8	When the required regulations have not been followed	H	mobile tower
6.1.9	Placed below the base plate to lend greater stability to the scaffolding	I	trestle working platform
6.1.10	A type of small wheel that can swivel; it usually has a brake on it	J	castor
		K	base plates
		L	bearing surface

(10 × 1) (10)

6.2 State FIVE safety factors that must be considered when erecting a builder's trestle working platform. (5)

6.3 In SANS 398-3:2005, strict guidelines are provided for moving a mobile tower.

State FIVE guidelines that must be adhered to when moving a mobile tower. (5)

6.4 List FIVE parts of an independent scaffold. (5)

[25]

TOTAL: 100

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